



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	34.8047	Longitude	-84.9023
Zone size (km)	10	Analysis date	2026-09-04

57

FOREST INTEGRITY SCORE / 100 Degraded Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.573	Good
NDVI Trend 3 years (Sentinel-2)	-0.063	Degrading
10-year trend (Landsat 8/9)	+0.274	Improving
40-year trend (Landsat 1984)	+0.288	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+0.72% / year	Amelioration
NDVI change 5 years	-0.035	Stable
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Growing forest (20-60 years)	Active growth phase
Above-ground biomass	~150.7 t/ha	High biomass (150-300 t/ha)
Canopy height	~18.3 m	Estimated from NDVI
AI Forest Type	Temperate Forest	Broadleaf/mixed, moderate climate
Sentinel-1 Radar (VV)	-9.406 dB	Dense forest (radar)
Forest cover	89.9%	Excellent
Fire pixels detected (FIRMS/year)	1731 pixels	~433 events estimated
Overall trend	degrading	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5	
Deforestation Alert	CAUTION — Light forest degradation	
Area change (3 years)	-239.3 ha	-2.15%
Annual rate (area/yr)	-79.8 ha/yr	Loss
Area change (10 years)	+10906.6 ha	+218132%

Annual rate 10yr (area)	+1090.7 ha/yr	Recovery
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SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	18	25
NDVI trend (3 years)	4	25
Forest cover	25	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	57	100

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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